Welcome to your next lesson: Wrap-up: JS basics.

I'm preparing your personalized lesson based on what you've learned, what's coming next, and how to help you lock in your skills.

Step by step, you're moving forward.

Ron, for this lesson we'll do a 6-round JavaScript basics challenge

Each round focuses on one core thing from unit 1:

Variables
Conditionals
Template literals
Strict equality & falsy
Arrays
Objects (plus a tiny integration at the end)

We'll go one round at a time, and you'll use the JavaScript editor (not the chat) to write and run code.

Let's start with Round 1: variables

In Python you might do:

Python
x = 5
print(x)
In JavaScript the same idea is:

Javascript
let x = 5;
console.log(x);

let or const = create a variable

= = assign a value

console.log(...) = like print(...) in Python

Your task for Round 1:

In the JavaScript editor:

Create a variable age with value 30 using let.
Reassign age to 31.
Log it with console.log(age);

Then Run, make sure it works, and when it looks good, hit Submit code.

Reply here once you've submitted, and I'll check it.

Javascript
let age = 30;
age = 31;
console.log(age);
Nice — Round 1 (variables) is perfect

You used let, reassigned, and logged the value. All good.

Round 2: conditionals & code blocks 🔀

In JavaScript, a basic if looks like:

Javascript
```
let score = 80;

if (score >= 70) {
  console.log("Pass");
} else {
  console.log("Fail");
}
```
The part inside { ... } is the code block that runs only if the condition is true.
This is similar to Python's if/else, just with {} and ;.
Your task for Round 2:

In the JavaScript editor:

Create let temperature = 15;
If temperature is less than 20, log: "It's cold"
Otherwise, log: "It's warm"

Then Run it, check the output, and when it looks right, hit Submit code.

Tell me when you've submitted, and we'll see how it looks. 📎

Javascript
```
let temperature = 35;
if (temperature < 20) {
    console.log("It's cold");
} else {
    console.log("It's warm");
}
```
That's exactly right ☑

You changed the value to 35, so "It's warm" is the correct output with your condition. Logic and syntax both look solid.

Round 3: template literals 🧩

In Python you might write:

Python
```
name = "Ron"
age = 31
print(f"{name} is {age} years old")
```
In JavaScript, the similar feature is a template literal:

Javascript
```
let name = "Ron";
let age = 31;

console.log(`${name} is ${age} years old`);
```
Use backticks `` ` ``
Insert variables with ${...}
Your task for Round 3:

In the JavaScript editor:

Make let name = "Ron";
Make let city = "Dallas"; (or any city you like)
Log exactly:
Ron lives in Dallas.
using a template literal (with ${name} and ${city}).

Run it, then Submit code, and tell me when it’s submitted. 🎯

Javascript
```
let name = "Ron";
let city = "Dallas";
console.log(`${name} lives in ${city}`);
```
Clean work again — template literal looks perfect, and the sentence is exactly right ☑

Round 4: strict equality and falsy values 🕹

Quick idea:

In JavaScript, === checks value and type (this is the one we want to use).
Some values are falsy — they behave like false in conditions: 0, "" (empty string), null, undefined, NaN, and false itself.
Non-empty strings like "hello" are truthy (treated like true in if).
Your task for Round 4:

In the JavaScript editor:

Create:

let message = "";
Write an if:

If message === "", log: "Message is empty".
Otherwise, log: "Message is not empty".
Then change message to "hello" and run again to see the different output.

When you’re happy with it, hit Submit code, and tell me you’ve submitted.

Javascript
```
let message = "";
if (message === "") {
    console.log("message is empty")
} else {
    console.log("message is not empy")
}
message = "hello";
if (message === "") {
    console.log("message is empty")
} else {
    console.log("message is not empy")
}
```
Nice — that’s exactly the behavior we wanted:

With message = "" → "message is empty"
With message = "hello" → "message is not empy" (tiny typo, but logic is correct)

And you used === correctly. 💯

Round 5: arrays and .length vs last index 📑

Quick reminder:

Javascript
```
let colors = ["red", "green", "blue"];

console.log(colors.length);   // 3
console.log(colors[0]);       // "red"  (first)
console.log(colors[2]);       // "blue" (last, index 2)
```
Arrays are 0-based.
Last valid index is length - 1.
Your task for Round 5 (in the JavaScript editor):

Create this array:

```
let nums = [10, 20, 30];
```
Log the length: console.log(nums.length);

Log the last element using the length (no hardcoded 2):

Hint: use nums[nums.length - 1]

Run it, then Submit code, and tell me when it’s submitted.

Javascript
```
let nums = [10, 20, 30];
console.log(nums.length);
console.log(nums[nums.length - 1]);
```
Beautiful – you used .length and length - 1 exactly right ☑

Output 3 and 30 is spot on.

Round 6 (last one): objects + a small integration 🧩

Think: Python dict vs JS object.

Python:

Python
```
user = {"name": "Ron", "age": 31}
print(user["name"])
```
JavaScript:

Javascript
```
let user = {
  name: "Ron",
  age: 31
};

console.log(user.name);      // dot notation
console.log(user["age"]);    // bracket notation
```
Dot notation: user.name
Bracket notation: user["name"] (useful when the key is in a variable)
Your task for Round 6 (in the JavaScript editor):

Create an object:

```
let settings = {
  theme: "dark",
  language: "en"
};
```
Create a variable:

```
let key = "theme";
```
Log the theme using bracket notation and the key variable (no hardcoded "theme" in the brackets):

It should log "dark".

Run it, then hit Submit code, and tell me when it’s submitted. 🎯

Javascript

```
let settings = {
  theme: "dark",
  language: "en"
};
let key = "theme";
console.log(settings[key]);
```

Nice, Ron — that’s all 6 rounds done, and all of them solid 🎯

Quick visual recap:

Idea	JavaScript	Python “equivalent”
Variable	`let age = 31;`	`age = 31`
Print/output	`console.log(x)`	`print(x)`
Template text	`` `Hi ${name}` ``	`f"Hi {name}"`
Conditionals	`if (x > 5) {}`	`if x > 5:`
Array/list	`[10, 20, 30]`	`[10, 20, 30]`
Last element	`arr[arr.length - 1]`	`arr[-1]` (Python shortcut)
Object/dict	`{ name: "Ron" }`	`{"name": "Ron"}`
Access key var	`obj[key]`	`d[key]`

You’ve locked in the core JS basics that everything else will sit on top of 🙌

Next step is to move on to the next lesson in A taste of TypeScript.

Before you go: is there one of these (variables, conditionals, template literals, arrays, or objects) you’d like a quick extra mini-example of, or are you good to move on?